Tianyan Gao
(Published as Yan Gao-Tian)
Email: ygaotian@ucla.edu

Education

University of California, Los Angeles | Statistics and Data Science **2025 – 2027**
GPA: 3.81
Diablo Valley College **2023 – 2025**
GPA: 3.93

Research Interests

Self-supervised representation learning for multimodal foundation models, especially structured predictive representations that support temporal, causal, and compositional reasoning.

Relevant Coursework & Self-study

- **Math & Statistics:** Calculus, Proof-based Linear Algebra, Differential Equations, Mathematical Statistics, Statistical Inference.
- **CS / ML:** Data Analysis, Data Structures, Machine Learning, Deep Learning
- **NLP / Advanced:** Stanford CS224N Natural Language Processing

Skills

- **Programming Languages:** Python, C++, R
- **Libraries and frameworks:** PyTorch**,** Pandas, NumPy, Matplotlib, PyQt, Transformers
- **Other:** Video / Photo editing, Sketching, 3D Modeling / Rendering / Rigging

Experience

Research Intern, UCLA NLP Group *October 2025 – April 2026*

- Ported VeRL reinforcement learning framework to AWS Trainium; adapted Ray distributed logic for Neuron Cores and implemented NxD workers to resolve hardware-specific compilation and execution bottlenecks.
- Adapted evaluation pipeline to align inference results on Trainium with CUDA-based official performance, establishing consistent and comparable experimental conditions.
- Co-authored **"OpenVLThinkerV2: A Generalist Multimodal Reasoning Model for Multi-domain Visual Tasks"** (under review at COLM 2026).

Research Intern, Renmin University of China *July 2025 – October 2025*

- Systematically categorized VLM hallucination patterns and proposed lightweight mitigation strategies.
- Built and trained a small-scale image captioning model from scratch to empirically validate hallucination sources.

Guided Research Project: Etiological Analysis of Lung Cancer *2023 - 2024*

- EDA on public lung cancer dataset (Pandas/Matplotlib): identified smoking/obesity as primary risk factors.

Vice-President, Cognitive Science & AI Club *2023 - 2024*

- Fostered an academic community for AI and Cognitive Science on campus, organizing multiple seminars on cutting-edge topics.

Technical Director, Animation Project *2022 - 2023*

- Developed a production-pipeline plugin using Python and Qt to streamline modeling, rigging, and animation workflows.
- Implemented non-photorealistic rendering (NPR) shaders to achieve stylized visual effects.

Computer Science Tutor *2024 - 2025*